Gell-Mann–Nishijima from the Galois Structure:

$Q = I_3 + Y/2$ algebraically derived

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Contents

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Status markers

[K] *Confirmed* — algebraically complete.
[B] *Established* — secured by verification script.
[S] *Speculative* — conjecture, not yet fully proved.

Abstract

The Gell-Mann–Nishijima relation $Q = I_3 + Y/2$ is derived from two already established Galois properties: (1) Hypercharge $Y = -1 + \frac{4}{3}\mathsf{NQR}_{13}$ from the Legendre symbol mod 13 (Doc. 346, Theorem A). (2) Isospin $I_3 = \pm\frac{1}{2}$ from the Su/Sd split via the jE7 projector (Doc. 344, Theorem E). The combination gives all four fermion charges $\{0, -1, +\frac{2}{3}, -\frac{1}{3}\}$ exactly, without free parameters. Theorem E [**[S]**] of Doc. 346 is thereby closed: **[B]**.

1 Background

Doc. 346 established the fermion classification from two Galois properties: QR/NQR bit (electric neutrality) and $N \bmod 3$ (colour charge). Theorem E [**[S]**] left open: does $Q = I_3 + Y/2$ also follow from Galois structure?

2 Theorem A — Hypercharge from QR/NQR

Theorem 2.1 (Hypercharge Y from Legendre symbol **[B]**).

$$Y = -1 + \frac{4}{3} \cdot \mathsf{NQR}_{13}, \qquad \mathsf{NQR}_{13} = \begin{cases} 0 & k \bmod 13 \in \mathsf{QR}_{13} \quad (\textit{leptons}) \\ 1 & k \bmod 13 \in \mathsf{NQR}_{13} \quad (\textit{quarks}) \end{cases}$$

gives $Y_{\text{lepton}} = -1$ *and* $Y_{\text{quark}} = +\frac{1}{3}$.

Proof. Leptons (QR orbits O_1, O_3): NQR $= 0 \Rightarrow Y = -1 + 0 = -1$ ✓
Quarks (NQR orbits O_2, O_4): NQR $= 1 \Rightarrow Y = -1 + \frac{4}{3} = \frac{1}{3}$ ✓ □ □

Connection to Doc. 346 [B]. The NQR bit is exactly the Legendre symbol $\left(\frac{k}{13}\right) = -1$ — the same property classifying electric neutrality in Doc. 346. Hypercharge and electric neutrality have the same algebraic source.

3 Theorem B — Isospin from Su/Sd split

Theorem 3.1 (Isospin I_3 from Su/Sd split **[B]**).

$$I_3 = \begin{cases} +\frac{1}{2} & \textit{Su sector (Doc. 344: } F \cdot jE7 = +F\textit{)} \\ -\frac{1}{2} & \textit{Sd sector (Doc. 344: } F \cdot jE7 = -F\textit{)} \end{cases}$$

The Su/Sd separation by the jE7 projector (Doc. 344, Theorem E) is exactly the doublet splitting of weak isospin. No new assumption — the jE7 projector was already established in the GALG/Furey construction.

4 Theorem C — Gell-Mann–Nishijima from Galois

Theorem 4.1 (Gell-Mann–Nishijima algebraically derived **[B]**). *$Q = I_3 + Y/2$ with I_3 from Theorem B and Y from Theorem A:*

Fermion	I_3	Y	$Q = I_3 + Y/2$	*Exp.*
Neutrino	$+\frac{1}{2}$	-1	0	0 ✓
Comparison values: PDG 2022 [6]; all four charges reproduced exactly.				
Electron	$-\frac{1}{2}$	-1	-1	-1 ✓
up quark	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{2}{3}$	$+\frac{2}{3}$ ✓
dn quark	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$ ✓

*All four values correct without free parameters **[B]**.*

5 Theorem D — Hypercharge from Legendre symbol (orbit level)

Orbit	$\left(\frac{k}{13}\right)$	Y
$O_1 = \{1, 3, 9\}$	$+1$ (QR)	-1
$O_3 = \{4, 10, 12\}$	$+1$ (QR)	-1
$O_2 = \{2, 5, 6\}$	-1 (NQR)	$+\frac{1}{3}$
$O_4 = \{7, 8, 11\}$	-1 (NQR)	$+\frac{1}{3}$

6 Summary and closure of Theorem E (Doc. 346)

Table 1: Results Doc. 347

Theorem	Content	Status
A	$Y = -1 + \frac{4}{3}$NQR from Legendre symbol	**[B]**
B	$I_3 = \pm\frac{1}{2}$ from jE7 projector (Su/Sd)	**[B]**
C	$Q = I_3 + Y/2$ fully from Galois structure	**[B]**
D	Y at orbit level from Legendre symbol	**[B]**

Theorem E [**[S]**] of Doc. 346 is closed. The Gell-Mann–Nishijima relation follows from two established Galois properties: the jE7 projector (Su/Sd split, Doc. 344) and the Legendre symbol mod 13 (QR/NQR split, Doc. 346). No new postulate.

7 What remains open: SU$(2)_L$ and CKM [S]

SU$(2)_L$ chirality. Chirality requires $\gamma_5 = e_1 \cdots e_6 \in$ Cl(6). The $\mathbb{Z}_2$ parity in GF$(27)^*$ separates particles from anti-particles ($k < 13$ vs. $k \geq 13$), not left-handed from right-handed. All four primitive orbits $O_1 \dots O_4$ are same-chiral. Needed: the bridge GF$(27)^* \to$ Cl$(6) \xrightarrow{\gamma_5}$ chirality. Begun in Docs. 341/344, not yet complete. **[S]**

CKM mixing angles. GF(27) carries a trace bilinear form $\langle a, b\rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(ab)$. The Cabibbo angle between generations would be the overlap $|\langle O_{2,\mathrm{Gen.1}}, O_{4,\mathrm{Gen.2}}\rangle|/(\|O_2\|\|O_4\|)$. Experimental: $\sin\theta_C \approx 0.225$; Galois candidate $\sin(\pi/14) \approx 0.222$ ($-1.2\,\%$). Three checked directions: (1) direct ξ-corrections: ratio ≈ 1685, no mechanism. (2) $\xi^{|k_i - k_j|}$ weighting: values $\sim 10^{-8}$, no mechanism. (3) Galois quotients: $2/9 \approx 0.222$ ($-1.4\,\%$), $3/13 \approx 0.231$ ($+2.4\,\%$), near but not forced. Full derivation requires its own document. **[S]**

Verification script

`pruef_347_gell_mann.py` — Theorems A–D, all assertions passed **[B]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions by Johann Pascher (ORCID 0009-0000-6518-4064).

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